

Grade 11 - Term 3

C1–C4 Practice Exercises and Solutions

Contents

1 Waiting for a Study-Room Opening [C4-5]

At a university library in Bogotá, study rooms become available at an average rate of one room every 5 minutes. Let T represent the waiting time, in minutes, until the next study room becomes available.

Questions/Tasks.

C3 Determine the probability that the next available study room appears between 2 and 7 minutes after the observation time. [4]

C4 - Exponential probability distribution [4]

1.1 Rate Parameter

The average rate is one available room every 5 minutes. Therefore, the exponential rate parameter is

$$\lambda = \frac{1}{5} = 0.20.$$

Thus, T has an exponential distribution with rate

$$\lambda = 0.20.$$

1.2 Exponential Probability Rule

For an exponential random variable, the probability that T lies between a and b is

$$P(a < T < b) = e^{-\lambda a} - e^{-\lambda b}.$$

In this problem,

$$a = 2, \quad b = 7, \quad \lambda = 0.20.$$

Probability region

1.3 Substitution

Substituting $a = 2$, $b = 7$, and $\lambda = 0.20$,

$$P(2 < T < 7) = e^{-(0.20)(2)} - e^{-(0.20)(7)}.$$

Therefore,

$$P(2 < T < 7) = e^{-0.4} - e^{-1.4}$$

1.4 Final Probability

Evaluating the expression gives

$$P(2 < T < 7) \approx 0.4259.$$

Thus, the probability that the next study room becomes available between 2 and 7 minutes after the observation time is

$$P(2 < T < 7) \approx 0.4259 = 42.59\%$$